

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm. Examination

University Theory Examination October/November 2016

PH215 PHARMACOLOGY I

Date: 22.11.2016, Tuesday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- | | | |
|------------|---|-----------|
| Q 1 | Attempt all questions. Each question is of one mark. | 20 |
| 1. | Which one of the following does not rely on topical drug delivery?
[A] Insulin pen
[B] Nasal Spray
[C] Nicotine patch
[D] Anti-dandruff Shampoo | |
| 2. | Pilocarpine may be used in the treatment of
[A] Abdominal Spasm
[B] Dryness of mouth
[C] Pheochromocytoma
[D] Hypertension | |
| 3. | Study of the drug effects at the population level is known as
[A] Pharmacoepidemiology
[B] Pharmacokinetics
[C] Pharmacogenomics
[D] Pharmacoeconomics | |
| 4. | Volume of distribution equals
[A] Urine concentration/plasma concentration
[B] Total amount of drug in the body/plasma concentration
[C] Urine drug concentration/plasma concentration
[D] Dose given/urine concentration | |
| 5. | G- Protein Coupled Receptors include all of the following EXCEPT
[A] Muscarinic Receptors
[B] Alpha Receptors
[C] Beta Receptors
[D] Nicotinic Receptors | |
| 6. | Which of the following reaction is categorized as phase I reaction:
[A] Oxidation
[B] Glucuronidation
[C] Acetylation
[D] Methylation | |

7. A pro-drug is
[A] A drug given to promote growth
[B] A drug given in its active form
[C] A drug given to prevent metabolism of another drug
[D] A drug given in its inactive form, requiring metabolism
8. A drug that binds to a cell receptor and affects a response is called
[A] An antagonist
[B] An agonist
[C] A receptor blocker
[D] A channel blocker
9. Regarding second messengers
[A] cAMP has no role in calcium homeostasis
[B] cAMP exerts most of its effects by stimulating cAMP-dependent protein kinases
[C] inhibition of adenylylcyclase results in increased cAMP
[D] phospholipase C is situated in the cell nucleus
10. A drug moves down its electrochemical gradient while cross plasma membrane in
[A] Active Transport
[B] Passive Diffusion
[C] Exocytosis
[D] Endocytosis
11. Which is the neurotransmitter present at the neuroeffector site of parasympathetic nervous system?
[A] Dopamine
[B] Acetylcholine
[C] Noradrenaline
[D] Adrenaline
12. Which of the following terminology is used to describe capacity of drug to cause foetal abnormalities when administered to the pregnant mother?
[A] Teratogenicity
[B] Mutagenicity
[C] Carcinogenicity
[D] Phototoxic reaction
13. Which of the following is an example of neuromuscular blocking agents?
[A] Doxazosin
[B] Propranolol
[C] D-tubocurarine
[D] Pilocarpine
14. Which of the following is used to treat organophosphorus poisoning?
[A] Pralidoxime
[B] Physostigmine
[C] Gallamine
[D] Phenylephrine
15. Which of the following drug is used to treat bronchial asthma?
[A] Albuterol
[B] Prazosin
[C] Atenolol
[D] Neostigmine
16. A drug that acts at prejunctional α_2 -adrenoceptors and is used to treat hypertension is
[A] Clonidine
[B] Methoxamine
[C] Metaproterenol

- [D] Dobutamine
17. Which of the following drug is contraindicated in the patients with Glaucoma
 [A] Pilocarpine
 [B] Rivastigmine
 [C] Atropine
 [D] Phenylephrine
18. Which body system is more vulnerable to the effects of local anesthetics?
 [A] Urinary
 [B] Reproductive
 [C] Nervous
 [D] Respiratory
19. Which of the following drug is used to prevent migraine attack?
 [A] Tamsulosin
 [B] Prazosin
 [C] Propranolol
 [D] Tiotropium
20. A drug used for diagnosis of myasthenia gravis would be expected to have improved neuromuscular function:
 [A] Donepezil
 [B] Edrophonium
 [C] Atropine
 [D] Echothipate

SECTION – II

Q 2 Attempt any TWO of the following

- A Define Pharmacokinetics. Write a note on factors affecting absorption of drug. 05
- B Define Pharmacodynamics. Write a note on G-Protein Coupled Receptor (GPCR) 05
- C Discuss the different phases of clinical trial. 05

Q 3 Attempt any TWO of the following

- A Differentiate: Competitive Antagonist and Non-Competitive Antagonist 05
- B Differentiate: Autonomic nervous system and somatic nervous system 05
- C Justify the use of Beta Blockers in the treatment of Hypertension and Angina pectoris 05

SECTION – III

Q 4 Attempt any FOUR of the following

- A Enlist routes of drug administration. Write merits and demerits of Intravenous route of administration 05
- B Classify Parasympatholytics. Explain adverse effects and therapeutic uses of atropine. 05

- C How plasma protein binding alter the drug effect? Explain with examples. 05
- D Explain phase II reactions of metabolism in detail. 05
- E Define clearance. Explain role of kidney in excretion of drugs. 05
- F Classify Local Anesthetics. Explain the therapeutic uses of them. 05
- Q 5 Attempt any FOUR of the following**
- A Classify Sympathomimetics. Explain Dale's vasomotor reversal. 05
- B Classify adverse effects of drug. Differentiate type A and type B ADR. 05
- C Explain pharmacokinetic drug interaction in detail. 05
- D Differentiate: D-tubocurarine and Succinylcholine 05
- E Define the following Terms:
1. Agonist
 2. Iatrogenic Diseases
 3. Volume of Distribution 05
 4. Drug tolerance
 5. Bioavailability
- F Classify parasymphomimetics. Explain the pharmacological actions and therapeutic uses of them. 05